

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

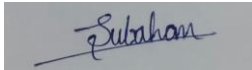
Total Marks: 50

Code of Conduct

I T.Subahan(192511121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of	20%	Covers all	Covers most	Covers only basic	Very limited

Concepts		expected sub-points completely	expected points	points	coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Cloud Computing:

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and processing power over the internet. It allows users to access these resources on demand without owning or maintaining physical infrastructure.

Four characteristics:

1. On-Demand Self-Service

- Users can access computing resources such as virtual machines or storage whenever they need them without requiring assistance from the service provider.
- **Difference from on-premises:** In traditional computing, IT staff must install, configure, and allocate hardware before it can be used.

2. Broad Network Access

- Cloud services are available over the internet and can be accessed from various devices, including laptops, smartphones, and tablets.
- **Difference from on premises:** Traditional systems are often limited to a company's local network or require special remote access.

3. Rapid Elasticity (Scalability)

- Cloud resources can be increased or decreased quickly based on demand. For example, additional storage or computing power can be added instantly during peak usage.
- **Difference from on premises:** Scaling traditional infrastructure requires purchasing, installing, and configuring additional hardware, which is time-consuming and expensive.

4. Measured Service (Pay-as-You-Go)

- Cloud providers monitor resource usage, and customers pay only for the resources they consume.
- **Difference from on premises:** Traditional computing requires large upfront investments in hardware and software regardless of actual usage, along with ongoing maintenance costs.

Cloud computing provides flexible, internet-based access to computing resources with features such as on-demand self-service, broad network access, rapid scalability, and pay-as-you-go pricing. These characteristics make it more cost-effective, scalable, and convenient than traditional on-premises computing.

2. Evolution of Cloud Computing:

Cloud computing evolved through several technological stages, from improvements in computer hardware to internet-based software services. Each stage contributed to the development of today's cloud platforms.

1. Hardware Evolution (Mainframes → Personal Computers)

- Early computing relied on large mainframe computers shared by many users.
- The introduction of personal computers (PCs) made computing affordable and accessible to individuals and businesses.
- Improvements in processors, memory, and storage increased computing power and reduced costs.

2. Networking Evolution

- The development of Local Area Networks (LANs) and Wide Area Networks (WANs) enabled computers to communicate and share resources.
- High-speed internet and broadband connections allowed data to be transferred quickly across long distances.

3. Virtualization Technology

- Virtualization allowed multiple **virtual machines (VMs)** to run on a single physical server.
- Resources such as CPU, memory, and storage could be shared efficiently among multiple users.

4. Internet and Software Evolution

- The internet became widely available, allowing applications to be delivered online.
- Web technologies evolved from static websites to interactive **web applications** and **Software as a Service (SaaS)**.
- Users no longer needed to install software locally; applications could be accessed through a web browser.

The combined evolution of hardware, networking, virtualization, and internet software made modern cloud computing possible by providing:

Scalable infrastructure through virtualization and large data centers.

On-demand access to computing resources via the internet.

Cost efficiency through shared infrastructure and pay-as-you-go pricing.

Global accessibility from any internet-connected device.

- Reliable and flexible services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

3. Role of Server Virtualization in Cloud Computing:

Server Virtualization

Server virtualization is the process of dividing one physical server into multiple virtual servers (Virtual Machines - VMs) using a hypervisor. Each VM has its own operating system and applications and works independently.

Role of Server Virtualization in Cloud Computing

- **Efficient Resource Utilization:** Multiple VMs share the same physical hardware, reducing idle resources.
- **Cost Reduction:** Fewer physical servers are required, lowering hardware, power, and maintenance costs.
- **Scalability:** New virtual servers can be created or removed quickly based on demand.
- **Isolation:** Each VM operates independently, improving security and reliability.
- **Easy Backup and Recovery:** Virtual machines can be cloned, migrated, and restored easily.
- **High Availability:** Workloads can be moved between servers with minimal downtime.

Virtualization (Enabling Technology): It is a technology that creates multiple virtual machines on a single physical server.

Cloud Computing (Service Model): It is a model that delivers computing resources over the internet.

Virtualization: Focuses on efficient use of hardware resources.

Cloud Computing: Focuses on providing services like IaaS, PaaS, and SaaS to users.

Virtualization: Runs within a data center or local environment.

Cloud Computing: Provides on-demand access to resources from anywhere through the internet.

4. Compare SOAP-based and REST-based web services:

SOAP (Simple Object Access Protocol):

- SOAP is a protocol used for exchanging structured information between applications.
- It follows strict standards and uses XML for communication.
- It is more secure and reliable, making it suitable for enterprise applications such as banking and healthcare.
- Uses a protocol-based communication model with strict rules.
- Exchanges data only in XML format.
- Slower because XML messages are larger and require more processing.
- Provides built-in security features such as WS-Security
- Best for enterprise applications that require high security, reliability, and transactions.

REST (Representational State Transfer):

- REST is an architectural style used to build web services.
- It is lightweight, flexible, and mainly uses HTTP methods (GET, POST, PUT, DELETE).
- It supports multiple data formats such as JSON, XML, and plain text.
- It is widely used in cloud computing, mobile applications, and web APIs.
- Uses an HTTP-based architectural style with standard HTTP methods.
- Exchanges data in JSON, XML, HTML, or plain text (JSON is most common).
- Faster because it usually uses lightweight JSON data.
- Uses HTTPS, OAuth, JWT, and other web security mechanisms.
- Best for cloud applications because it is lightweight, scalable, faster, and easy to integrate with web and mobile services.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Suitable Real-World

Example for Each Model

IaaS (Infrastructure as a Service):

- Provides virtualized computing resources such as servers, storage, and networking over the internet.
- **User Responsibility:** Manage the operating system, applications, and data.
- **Provider Responsibility:** Manage the physical hardware and virtualization.
- **Example:** Amazon EC2 (AWS)

PaaS (Platform as a Service):

- Provides a platform for developers to build, test, and deploy applications without managing the underlying infrastructure.
- **User Responsibility:** Manage only the application and its data.

- **Provider Responsibility:** Manage servers, operating system, runtime, and middleware.
- **Example: Google App Engine**

SaaS (Software as a Service):

- Provides ready-to-use software applications over the internet.
- **User Responsibility:** Use the software.
- **Provider Responsibility:** Manage the application, infrastructure, updates, and security.
- **Example: Google Workspace (Gmail, Google Docs)**

XaaS (Anything as a Service):

- Refers to any IT service delivered over the internet, including IaaS, PaaS, SaaS, Storage as a Service, Database as a Service, and more.
- **User Responsibility:** Use the required service.
- **Provider Responsibility:** Manage and deliver the service.
- **Example: Microsoft Azure**

Differences:

IaaS:

- Provides infrastructure (servers, storage, networking).
- User manages OS and applications.
- **Example:** Amazon EC2.

PaaS:

- Provides a development platform.
- User develops and deploys applications.
- **Example:** Google App Engine.

SaaS:

- Provides ready-to-use software.
- User only uses the application.
- **Example:** Google Workspace.

XaaS:

- Provides any IT service over the internet.
- Includes IaaS, PaaS, SaaS, and many other services.
- **Example:** Microsoft Azure.